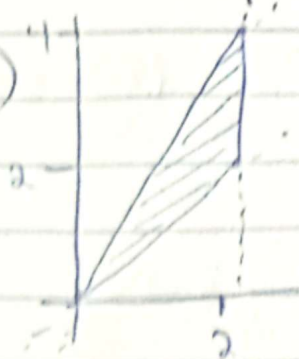


Q1 a)



Intersection Points:

$$\rightarrow (0,0)$$

$$\rightarrow (2,2)$$

$$\rightarrow (2,4)$$

Scalene Triangle

b) 1. $dy dx : \int_0^2 \int_n^{2n} n^2 y + \sin ny \, dy \, dx$

2. $dx dy : \int_0^4 \int_{ny/2}^{2y} n^2 y + \sin(ny) \, dx \, dy$
 $+ \int_2^4 \int_{y/2}^2 n^2 y + \sin(ny) \, dx \, dy$

c) 1. is easier since it is just 1 double integral

d) (Integrating 1.) $\int_0^2 \int_n^{2n} n^2 y + \sin ny \, dy \, dx$

$$= \int_0^2 \left[\frac{n^2 y^2}{2} - \frac{\cos ny}{n} \right]_n^{2n} dx$$

$$= \int_0^2 \left[\frac{4n^4}{2} - \frac{\cos(2n^2)}{n} - \frac{n^4}{2} + \frac{\cos n^2}{n} \right] dx$$

$$= \int_0^2 \frac{3}{2} n^4 + \frac{\cos n^2 - \cos 2n^2}{n} \, dx$$

$$= \left[\frac{3}{2x^3} n^4 + \frac{1}{2} \right] \frac{\cos u - \cos 2u}{n} du \text{ where } u=n^2$$

$$= \frac{48}{5} + \frac{1}{2} \int \frac{\cos u - \cos 2u}{u} du$$

$$\begin{aligned} \text{e) Area of region} &= \frac{1}{2}(2 \times 4) - \frac{1}{2}(2 \times 2) \\ &= 4 - 2 \\ &= 2 \end{aligned}$$

$$\therefore \text{Average} = \frac{1}{2} \left(\frac{48}{5} + \frac{1}{2} \int \frac{\cos u - \cos 2u}{u} du \right)$$

$$= \frac{48}{5} + \frac{1}{4} \int \frac{\cos u - \cos 2u}{u} du$$

Q2

a) Polar Coordinates are better because $x^2 + y^2$ can be easily simplified to r^2 , and the region is circular, hence the limits from the origin is easier

$$f(r, \theta) = 4000e^{-0.01r^2}$$

$$25r^2 \leq 49$$

$$\begin{aligned} \text{where } 0 \leq r \leq \sqrt{49} &, \quad 0 \leq \theta \leq 2\pi \\ &= 0 \leq r \leq 7 \end{aligned}$$

$$\text{b) } P = \int_0^{2\pi} \int_0^7 4000e^{-0.01r^2} r dr d\theta$$

$$c) \int_0^{2\pi} \int_0^1 4000 r e^{-0.02r} dr d\theta$$

$$\text{let } u = -0.02r, \quad du = -0.02 dr$$

$$\therefore r dr = \frac{du}{-0.02}$$

$$0 \leq u \leq -0.49$$

$$\therefore \int_0^{2\pi} \frac{4000}{-0.02} \int_0^{-0.49} e^u du d\theta$$

$$= \int_0^{2\pi} \frac{4000}{-0.02} [e^u]_0^{-0.49} d\theta$$

$$= \int_0^{2\pi} \frac{4000}{-0.02} (e^{-0.49} - 1) d\theta$$

$$= \frac{4000 (e^{-0.49} - 1) \times 2\pi}{-0.02}$$

$$= -400,000\pi (e^{-0.49} - 1)$$

$$= 400,000\pi (1 - e^{-0.49})$$

$$\approx \boxed{486835}$$

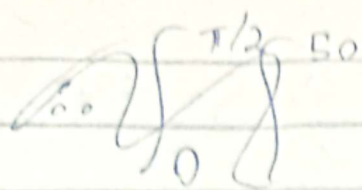
Q3)

$$a) \quad x = r \cos \theta, \quad y = r \sin \theta$$

$$x^2 + y^2 = r^2$$

$$\therefore z_f = \frac{1}{4} (r \cos \theta + 2 r \sin \theta)$$

$$z_c = 15 + \frac{r^2}{100}$$



$x > 0, y > 0$, $\therefore$ first quadrant
which is $0 \leq \theta \leq \pi/2$

and $r \leq 50$

Hence
$$\int_0^{\pi/2} \int_0^{50} \int_0^{1 + \frac{r^2}{100}} \frac{r}{4} (\cos \theta + 2 \sin \theta) r \, dz \, dr \, d\theta$$

$$= \int_0^{\pi/2} \int_0^{50} \left[\frac{1}{2} r^2 + \frac{r^4}{400} - \frac{r^3}{12} (\cos \theta + 2 \sin \theta) \right]_0^{50} d\theta$$

$$= \int_0^{\pi/2} \left[\frac{1}{2} r^2 + \frac{r^4}{400} - \frac{r^3}{12} (\cos \theta + 2 \sin \theta) \right]_0^{50} d\theta$$

$$= \int_0^{\pi/2} \left[18750 + 15625 - \frac{125000}{12} (\cos \theta + 2 \sin \theta) \right] d\theta$$

$$= \int_0^{\pi/2} \left[34375 - \frac{31250}{3} (\cos \theta + 2 \sin \theta) \right] d\theta$$

$$= \left[\theta (34375) - \frac{31250}{3} (\sin \theta - 2 \cos \theta) \right]_0^{\pi/2}$$

$$= \frac{34375\pi}{2} - \left(\frac{31250}{3} + \frac{62500}{3} \right)$$

$$= \boxed{22746.5}$$

b) $\frac{\partial z}{\partial x} = \frac{x}{50}, \quad \frac{\partial z}{\partial y} = \frac{y}{50}$

$$\therefore S = \sqrt{1 + \frac{x^2}{50^2} + \frac{y^2}{50^2}}$$

$$S = \sqrt{1 + \frac{r^2}{50^2}}$$

$$6. \quad S = \int_0^{\pi/2} \int_0^{50} \sqrt{1 + \frac{r^2}{2500}}$$

let $u = \frac{r^2}{2500}$, $du = \frac{2r}{2500} dr$, $\therefore dr = \frac{1250 du}{1}$, $1 \leq u \leq 2$

$$\therefore S = \int_0^{\pi/2} \int_1^2 \sqrt{u} \cdot 1250 du d\theta$$

$$S = \int_0^{\pi/2} \left[\frac{2\sqrt{u}^3}{3} \right]_1^2 d\theta$$

$$= \frac{1250\pi}{3} (2\sqrt{2} - 1)$$

$$\approx \boxed{27394.48}$$

24) $P(x, y, z) = 1 - x^2 - y^2 - z^2$

a) $P(x, y, z) \geq 0$

$\therefore x^2 + y^2 + z^2 \leq 1$ $\therefore Q$ is a sphere with $r=1$
origin $(0,0,0)$

b) The triple integral represents the sum of the total energy within the region Q

$$c) \int_0^{2\pi} \int_0^{\pi} \int_0^1 (1 - r^2) r^2 \sin \phi dr d\phi d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi} \left[\sin \phi \left(\frac{r^3}{3} - \frac{r^5}{5} \right) \right]_0^1 d\phi d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi} \frac{2 \sin \phi}{15} d\phi d\theta$$

$$= \frac{2}{15} \int_0^{2\pi} \frac{4}{15} [-\cos \phi]_0^{\pi} d\phi$$

$$= \frac{2}{15} \int_0^{2\pi} \frac{4}{15} d\phi$$

$$= \frac{4}{15} (2\pi) = \boxed{\frac{8\pi}{15}}$$

$$\approx \boxed{1.676}$$

Q5) $x^2 + y^2 + z^2 = 16$, $z = k \sqrt{x^2 + y^2}$

a) $r^2 = x^2 + y^2$, $\therefore r^2 + z^2 = 16$ &
and $z = kr$

$\therefore z = \sqrt{16 - r^2}$ (Dome)
and $z = kr$ (Cone)

$$kr = \sqrt{16 - r^2}, \therefore k^2 r^2 = 16 - r^2$$

$$\therefore r^2 = \frac{16}{k^2 + 1}$$

$$\therefore r = \frac{4}{\sqrt{k^2 + 1}}$$

$$\therefore 0 \leq \phi \leq 2\pi, 0 \leq r \leq \frac{4}{\sqrt{k^2 + 1}}, kr \leq z \leq \sqrt{16 - r^2}$$

b) $\int_0^{2\pi} \int_0^{\frac{4}{\sqrt{k^2 + 1}}} \int_{kr}^{\sqrt{16 - r^2}} r dz dr d\phi$

$$= \int_0^{2\pi} \int_0^{\frac{4}{\sqrt{k^2 + 1}}} r (\sqrt{16 - r^2} - kr^2) dr d\phi$$

$$= \int_0^{2\pi} \left[4\pi \frac{(16 - r^2)^3}{3} - \frac{kr^3}{3} \right] \frac{1}{\sqrt{k^2+1}} d\phi$$

$$= \frac{1}{3} \int_0^{2\pi} \left(\frac{4}{3} 16 \cdot 64 \left(1 - \frac{k}{\sqrt{k^2+1}} \right) \right) d\phi$$

$$= \left(\frac{1}{3} \right) (2\pi) \left(\right)$$

$$= \frac{4288\pi}{3} \left(1 - \frac{k}{\sqrt{k^2+1}} \right) \pi$$

c) $x_{av} = y_{av} = 0$ (Origin, cylinder)

$$z_{av} = \frac{1}{V M} \iiint z dV$$

$$\therefore z_{av} = \frac{\frac{64\pi}{k^2+1}}{V}$$

2b) $x = r \sin \phi \cos \theta$, $y = r \sin \phi \sin \theta$, $z = r \cos \phi$

$$r^2 = x^2 + y^2 + z^2, \quad dV = r^2 \sin \phi dr d\phi d\theta$$

a) $\therefore \rho(x, y, z) = k \sqrt{x^2 + z^2}$
where $a^2 \leq x^2 + z^2 \leq b^2$

b) $V = \int_0^{2\pi} \int_0^{\pi/4} \int_a^b r^2 \sin \phi dr d\phi d\theta$

Q2

```
import numpy as np
def population_density(r):
    return 4000 * np.exp(-0.01 * r**2)
def approximate_population(n_r, n_theta):
    r_vals = np.linspace(0, 7, n_r)
    theta_vals = np.linspace(0, 2 * np.pi, n_theta)
    dr = 7 / (n_r - 1)
    dtheta = (2 * np.pi) / (n_theta - 1)
    total_pop = 0
    for i in range(len(r_vals) - 1):
        r_mid = (r_vals[i] + r_vals[i+1]) / 2
        for j in range(len(theta_vals) - 1):
            total_pop += population_density(r_mid) * r_mid * dr * dtheta
    return total_pop
exact = 400000 * np.pi * (1 - np.exp(-0.49))
for grid in [10, 100, 500]:
    approx = approximate_population(grid, grid)
    print(f'Grid {grid}x{grid}: Approx = {approx:.2f}, Error = {abs(exact-approx):.4f}')
```

```
Grid 10x10: Approx = 487414.69, Error = 626.6633
Grid 100x100: Approx = 486793.20, Error = 5.1714
Grid 500x500: Approx = 486788.23, Error = 0.2035
Press any key to continue . . .
```

Q3

```
import numpy as np

R_max = 50
theta_max = np.pi / 2
N = 1000
r = np.linspace(0, R_max, N)
theta = np.linspace(0, theta_max, N)
dr = r[1] - r[0]
dtheta = theta[1] - theta[0]
R, T = np.meshgrid(r + dr/2, theta + dtheta/2)
dA = R * dr * dtheta
floor = 0.25 * (R * np.cos(T) + 2 * R * np.sin(T))
ceiling = 15 + (R**2 / 100)
volume = np.sum((ceiling - floor) * dA)
dz_dx = (R * np.cos(T)) / 50
dz_dy = (R * np.sin(T)) / 50
integrand_sa = np.sqrt(1 + dz_dx**2 + dz_dy**2)
surface_area = np.sum(integrand_sa * dA)
print(f'Approximate Volume: {volume:.2f} ft^3')
print(f'Approximate Ceiling Surface Area: {surface_area:.2f} ft^2')
```


Q3

```
C:\Windows\system32\cmd.exe
Grid 10x10: Approx = 487414.69, Error = 626.6633
Grid 100x100: Approx = 486793.20, Error = 5.1714
Grid 500x500: Approx = 486788.23, Error = 0.2035
Press any key to continue . . .
```

Q4

```
import numpy as np
np.random.seed(42)

n_samples = 1_000_000
box_volume = 8 # (2*2*2 cube)
points = np.random.uniform(-1, 1, (n_samples, 3))
r_sq = np.sum(points**2, axis=1)
density = np.where(r_sq <= 1, 1 - r_sq, 0)
total_energy_approx = box_volume * np.mean(density)

print(f'Approximated Total Energy: {total_energy_approx:.5f}')
print(f'Exact Value: {8 * np.pi / 15:.5f}')
```

```
C:\Windows\system32\cmd.exe
Approximated Total Energy: 1.67751
Exact Value: 1.67552
Press any key to continue . . .
```

Q5

```
import numpy as np

def approximate_tank(k=1.0, n_samples=1_000_000):
    r_max = 4 / np.sqrt(k**2 + 1)
    z_min = 0
    z_max = 4
    x = np.random.uniform(-r_max, r_max, n_samples)
    y = np.random.uniform(-r_max, r_max, n_samples)
    z = np.random.uniform(z_min, z_max, n_samples)
    r_sq = x**2 + y**2
    r = np.sqrt(r_sq)
    inside_sphere = (r_sq + z**2 <= 16)
    above_cone = (z >= k * r)
    is_liquid = inside_sphere & above_cone
    box_vol = (2 * r_max) * (2 * r_max) * (z_max - z_min)
    vol_approx = box_vol * np.mean(is_liquid)
    z_cm_approx = np.mean(z[is_liquid])

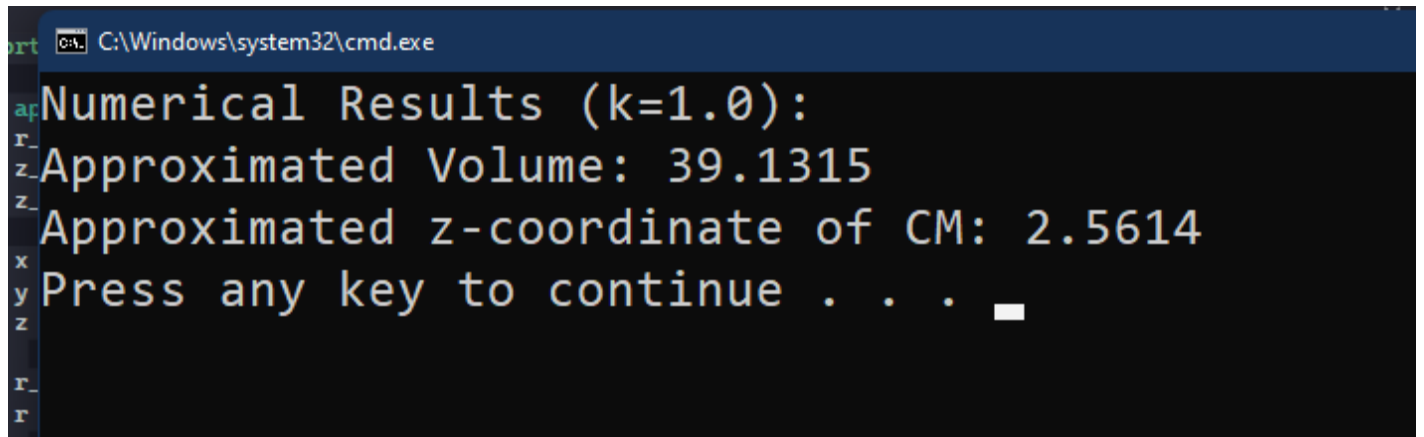
    return vol_approx, z_cm_approx
```

```

k_val = 1.0
vol, z_cm = approximate_tank(k=k_val)

print(f'Numerical Results (k={k_val}):')
print(f'Approximated Volume: {vol:.4f}')
print(f'Approximated z-coordinate of CM: {z_cm:.4f}')

```



```

C:\Windows\system32\cmd.exe
Numerical Results (k=1.0):
Approximated Volume: 39.1315
Approximated z-coordinate of CM: 2.5614
Press any key to continue . . .

```

Q6

```
import numpy as np
```

```
a, b, k = 1.0, 2.0, 1.0
```

```
n = 200
```

```
r = np.linspace(a, b, n)
```

```
theta = np.linspace(0, 2 * np.pi, n)
```

```
phi = np.concatenate([np.linspace(0, np.pi/4, n//2), np.linspace(3*np.pi/4, np.pi, n//2)])
```

```
dr = (b - a) / n
```

```
dtheta = (2 * np.pi) / n
```

```
dphi = (np.pi / 4) / (n // 2)
```

```
R, Theta, Phi = np.meshgrid(r, theta, phi, indexing='ij')
```

```
dV = (R**2) * np.sin(Phi) * dr * dtheta * dphi
```

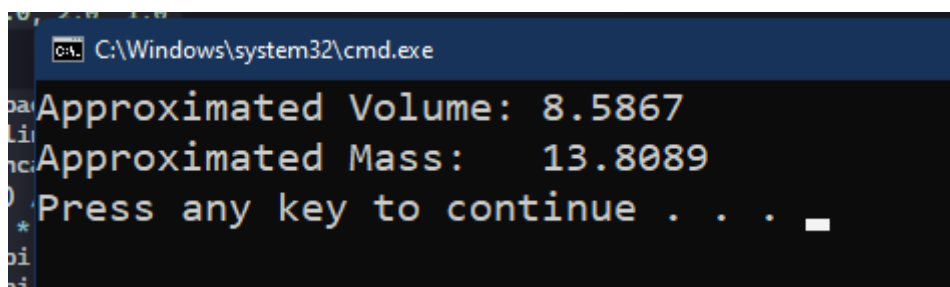
```
rho = k * R
```

```
volume = np.sum(dV)
```

```
mass = np.sum(rho * dV)
```

```
print(f'Approximated Volume: {volume:.4f}')
```

```
print(f'Approximated Mass: {mass:.4f}')
```



```

C:\Windows\system32\cmd.exe
Approximated Volume: 8.5867
Approximated Mass: 13.8089
Press any key to continue . . .

```